



DRAFT — NOT FOR SUBMISSION OR PARTICIPANT USE
Resolve all OPEN DECISION items and obtain UCF approval before use.

Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

Principal Investigator: **[OPEN DECISION: confirm UCF PI of record]**

Department/School: **[OPEN DECISION]**

Telephone: **[OPEN DECISION]**

Email: **[OPEN DECISION]**

Faculty Advisor: **[OPEN DECISION: required if the PI is a student]**

Version Number/Date: Version 1 / **[OPEN DECISION: submission date]**

1.0 Version History

Version	Date	Summary
1	[submission date]	Initial submission.

2.0 Study Summary

This laboratory study evaluates how two-step gaze-contingent spoken guidance and a participant-derived self-similar synthetic voice affect visual-search performance, workload, trust, and experience in mixed reality. Adults will remain seated in a swivel chair while wearing an HTC Vive Focus Vision headset and search 56 virtual colored shapes distributed across eight vertical planes surrounding them at 45-degree intervals. The system will provide spoken guidance under four within-participant conditions formed by gaze-contingent versus noncontingent guidance and a generic versus participant-derived self-similar voice.

Participants will provide approximately 1–2 minutes of speech for Instant Voice Cloning (IVC). The recording will be uploaded to an institutionally approved third-party speech service only after consent. The resulting voice will be used only to synthesize study prompts for that participant, will not be shared or published, and will be deleted under the approved deletion procedure. Eye-gaze telemetry, task events, performance, questionnaires, and limited demographics will be recorded under a study ID. No eye-camera images, room video, webcam video, or face video will be saved.

Protocol proposal: One in-person session of approximately **[OPEN DECISION: proposed 75–90 minutes]**, including consent, screening, voice recording, headset setup/calibration, up to eight nonanalyzed practice trials, four 16-trial experimental blocks (64 analyzed trials), questionnaires, breaks, debriefing, cleanup, and compensation if offered.

Study-readiness gate: The current Unity build implements a 14-round, front-facing bookshelf task with alternating “gaze-aware”/gaze-unaware rounds and a fixed generic voice. It does not implement the planned eight-plane 360-degree environment, 64-trial four-block schedule, Williams counterbalancing, participant-scoped voice playback, matched hint opportunities, two-step coarse/fine guidance policy, post-block questionnaires, or all required audit events. Voice recording, provider upload, prompt generation, and deletion will occur through a separate approved staff-controlled workflow, not in the headset application. No participant enrollment or research data collection will begin until the Unity release build and operational workflow together match this approved protocol and are verified with non-participant test data.

3.0 Background

Gaze can provide moment-to-moment information about what a person is inspecting in a visual-search task. Two-Step Gaze Guidance separates rapid approximate-direction guidance from fine target acquisition; related XR research also shows that spatial audio, visual cueing, and gaze-contingent feedback can support out-of-view search. This study adapts that architecture: a coarse spatialized utterance first orients the participant toward the target plane, and a fine stage uses angular gaze-target proximity to select discrete warmer/colder feedback. A voice that resembles the participant may change perceived familiarity, comfort, trust, or reliance. The study therefore manipulates guidance contingency and voice similarity independently to estimate their separate and combined effects.

Eye-gaze hover and dwell events generated by the XR Interaction Toolkit are interaction proxies. They are not treated as clinically validated fixations. Angular transitions are not treated as validated saccades. The research will use neutral, implementation-accurate terminology in reports unless a validated classification pipeline is added through an approved modification.

Community Advisory Board Input: N/A. This minimal-risk laboratory usability study does not currently use a community advisory board. **[OPEN DECISION: confirm with UCF.]**

4.0 Objectives

Primary scientific objectives:

- Estimate the effect of gaze-contingent two-step guidance versus noncontingent general prompts on visual-search time.
- Estimate whether self-similar versus generic voice changes the effect of gaze contingency on the pre-specified primary experience/performance outcome.

Secondary objectives:

- Estimate the effect of self-similar versus generic voice on workload, trust/helpfulness, comfort, and perceived voice similarity.
- Characterize wrong captures, head-yaw travel, target-plane latency, plane revisits, hint-response latency, gaze-hover/dwell inspection proxies, technical failures, and adverse symptoms.
- Evaluate feasibility and data quality while completing the simulation-based sensitivity/power analysis required before preregistration and confirmatory enrollment.

Primary outcome: search time in seconds from the logged `search\_onset` event, when all 56 objects are available, through correct gaze-dwell capture, including time spent on preceding wrong captures. H1 predicts shorter search time under gaze-contingent guidance. The guidance-by-voice interaction on search time is the second focal contrast and is directional but secondary because prior evidence for the interaction is limited. Holm's sequential procedure will control multiplicity across these two focal tests. **[OPEN DECISION: complete simulation-based power/sensitivity analysis and confirm whether this hierarchy is adequately powered before preregistration.]**

Secondary outcomes include wrong captures, first-attempt success, head-yaw travel, target-plane latency, plane revisits, hint-response latency, raw NASA-TLX ratings, guidance/voice manipulation checks, trust/helpfulness/reliance/control ratings, symptom ratings, valid gaze availability, gaze-hover/dwell inspection proxies, and completion/withdrawal.

5.0 What is Being Measured or Evaluated?

The study measures task performance, interaction telemetry, subjective workload, perceived assistance quality, perceived voice similarity, trust, comfort, prior XR/voice-AI experience, and adverse symptoms. The study does not diagnose health conditions and does not use eye tracking as a medical measure.

The implemented telemetry currently includes frame time, gaze origin/direction/orientation, per-eye position where available, eye openness where available, provider-reported gaze point, hovered object metadata, target status, dwell progress, target identity, application state, ray visibility, and a custom blink proxy. Discrete logs include task and hover/capture events and summary metrics.

Protocol proposal: Add calibration result, tracking validity/availability, explicit search-display onset, hint request/playback/failure, block and counterbalancing assignment, voice condition and non-identifying voice alias,

and cleanup/deletion result. Preserve raw values and calculate derived measures in version-controlled analysis code.

Clinical trial/clinical investigation: No. The study evaluates behavioral interaction with a mixed-reality research application and does not evaluate an FDA-regulated drug, biologic, or medical device for diagnosis or treatment. **[UCF must confirm this classification.]**

6.0 Drugs and Medical Devices (If not a drug or medical device study, check N/A and skip to the Endpoints section)

N/A. The HTC Vive Focus Vision and computer are used as research apparatus for a behavioral task, not as investigational medical devices. No drug, biologic, IDE, or abbreviated IDE is involved. **[UCF must confirm.]**

7.0 Endpoints (clinical trials only—if not a clinical trial, check N/A and skip to the Number of Subjects section)

N/A. This is not proposed as a clinical trial or clinical investigation. Behavioral outcomes are described under Objectives and Data Analysis Plan.

8.0 Number of Subjects

LOCAL: Target **48 complete participant datasets**. **[OPEN DECISION: enrolled maximum after power/sensitivity analysis and anticipated screening/technical attrition; distinguish enrolled from completed/analyzed.]**

ALL SITES: Same as local; this is currently a single-site UCF study.

The application will state anticipated screening failures and technical/data-quality attrition. Participants whose data are excluded remain enrolled participants and will be reported in the disposition summary.

9.0 Inclusion/Exclusion Criteria

Inclusion criteria:

- Age 18 years or older.
- Able to provide informed consent in English.
- Able to understand spoken English study instructions and prompts.
- Able to wear the headset, remain seated, and safely turn in a swivel chair to search the surrounding field, with or without ordinary vision correction compatible with the headset.
- Willing to provide a short recording of their own voice and authorize creation of a study-limited synthetic voice.

Proposed exclusion criteria, subject to PI/UCF confirmation:

- History of photosensitive seizure/epilepsy or a clinician's instruction to avoid immersive displays.
- Current severe dizziness, nausea, migraine, eye pain/infection, or other condition that makes headset use unsafe or intolerable.
- Inability to obtain a minimally usable eye-tracking calibration after the pre-specified retry procedure. This is a technical eligibility criterion, not a health diagnosis.
- Inability or unwillingness to complete required voice recording/cloning.
- Prior participation in this experiment or prior knowledge of the deterministic target schedule that could invalidate the task.

Pregnancy is not an automatic exclusion in this draft. **[OPEN DECISION: UCF/PI must confirm whether any additional scientifically justified criteria are needed.]**

Vulnerable/special populations: Children, prisoners, and adults unable to consent will not be enrolled. Recruitment will not target pregnant persons, economically or educationally disadvantaged persons, or UCF employees/students under the direct authority of the recruiting investigator. Adults over 65 are not excluded solely because of age.

Additional safeguards: Recruitment and consent will emphasize voluntariness. If students are recruited, instructors will not know who participates until grades are final, and any course-credit pathway will include an equivalent alternative assignment.

10.0 Recruitment Methods

Recruitment may use IRB-approved email, flyer, research-participation portal/SONA listing, and word-of-mouth referral that directs interested people to the approved materials. **[OPEN DECISION: select exact methods in Huron and remove unused methods.]**

Materials will disclose the in-person laboratory visit, mixed-reality headset, eye tracking, short voice recording and synthetic voice creation, estimated duration, basic eligibility, compensation if any, and researcher contact. Recruitment will not claim guaranteed benefit. The team will not recruit before approval and will not use unapproved social-media direct messaging.

Potential participants may contact the study team or open an approved screening form. Contact and screening records will be stored separately from research data. Recruitment will avoid undue influence from supervisory, employment, instructional, or grading relationships.

11.0 Consent Process

An IRB-approved adult consent form will be used. A trained study-team member will conduct consent in a private lab area before screening procedures that collect research information, voice recording, headset fitting, or data collection. The researcher will explain the study purpose, procedures, eye tracking, voice cloning, third-party processing, risks, compensation, confidentiality limits, voluntary nature, and withdrawal options. The participant will have time to read the form, ask questions, and decide without pressure.

The researcher will check comprehension with neutral questions about voluntariness, voice use, third-party upload, withdrawal, and whom to contact. Incorrect responses will be reviewed; the person will not enroll unless they demonstrate understanding and provide consent.

Electronic consent: **[OPEN DECISION: paper signature versus UCF-approved electronic consent; if electronic, specify platform, identity verification, copy delivery, and signature workflow in Huron.]**

Documentation of consent: A signed and dated consent will be obtained unless UCF grants a waiver of documentation. The participant will receive a copy. The research record will note the consent version, date/time, consenting team member, and whether voice-cloning authorization was provided. A separate voice-authorization initial line is included in the draft consent for clarity; UCF must approve its use.

Waiver/alteration of consent: None proposed.

Deception/incomplete disclosure: None proposed. Consent will fully disclose eye tracking, creation/use of the participant-derived voice, third-party processing, risks, and the fact that different assistance/voice configurations are compared. To reduce expectancy bias, condition labels, the live-gaze rule, and directional hypotheses will not be emphasized before outcome measurement; no false information will be provided. The post-participation information explains the comparison. **[OPEN DECISION: UCF must confirm whether this neutral description constitutes ordinary hypothesis withholding or incomplete disclosure requiring HRP-509.]**

Language: English only. **[OPEN DECISION: justify scientifically and confirm with UCF; otherwise translate materials and add qualified language support.]**

12.0 HIPAA (From medical records only. If data is not coming from medical records, check N/A and skip to the FERPA section)

N/A. The study does not access medical records, covered-entity records, or protected health information. Safety screening is collected directly from participants and is limited to present eligibility/safety needs.

13.0 FERPA (From educational records only. If data is not coming from educational records, check N/A and skip to the Study Procedures section)

N/A. The study does not access educational records. If a research-participation portal or course credit is used, participation/credit administration will be handled under the approved separation and alternative-assignment safeguards without importing grades or educational records into the research dataset. **[OPEN DECISION: confirm recruitment/credit plan.]**

14.0 Study Procedures (after the consent process)

Protocol proposal: The following numbered procedure is intended study behavior and is not participant-ready until every study-readiness gate and open decision in this protocol is resolved, implemented, verified with non-participant data, and approved by UCF.

1. Confirm consent and assign the next sequential study ID, containing no name, initials, or contact information. Map the enrollment sequence to the frozen counterbalancing schedule.
2. Complete eligibility and current-symptom screening. A person who is ineligible will not begin headset or voice procedures and will receive any compensation required by the approved plan, if compensation is offered.
3. Record approximately 1–2 minutes of the participant reading the approved neutral script in a private room using an institution-managed encrypted workstation. The file name will contain only the study ID and session/run identifier. Automatic cloud backup/synchronization will be disabled for the temporary recording location.
4. Upload the recording over an encrypted TLS connection to ElevenLabs IVC or the final institution-approved provider. Before collection, the research account will have model-improvement sharing disabled. The clone will use a non-identifying alias, will remain private, and will never be published or shared to a voice library. Provider access will be limited to designated staff using individual least-privilege accounts, multi-factor authentication when supported, and revocable credentials.
5. On the approved staff-controlled workstation, generate only the pre-enumerated study prompt library needed for that participant and record each generation in the restricted provider/deletion log. Copy the prepared audio clips to the headset under the study ID. The headset release build will contain no provider credential, make no provider API call, and accept no free-form synthesis text. Provider interfaces will not be used to generate speech outside the approved prompt library.
6. Fit and sanitize the HTC Vive Focus Vision, explain passthrough, adjust fit/vision correction, seat the participant at the marked origin in a swivel chair, establish a clear rotational area, and perform manufacturer/system eye calibration. Validate gaze at multiple directions spanning the eight-plane field. Record calibration/validation result and retry count; abort startup if the approved criterion is not met.
7. Complete standardized nonanalyzed practice with both guidance policies and both voices. Practice continues until the participant correctly completes three consecutive targets without intervention, up to eight trials.
[OPEN DECISION: freeze the handling rule if the participant does not meet the criterion.]
8. Complete four 16-trial blocks representing gaze-contingent/generic voice (C), gaze-contingent/self-similar voice (D), noncontingent/generic voice (A), and noncontingent/self-similar voice (B), for 64 analyzed trials. A four-sequence balanced Latin square (Williams design) assigns A-B-D-C, B-C-A-D, C-D-B-A, or D-A-C-B. A pre-generated randomized allocation schedule assigns 12 of the 48 complete datasets to each sequence; participant ID reveals the next assignment so the experimenter cannot select it. Within every condition, the target appears exactly twice on each of the eight planes. Shape, color, within-plane position, initial visibility, and layout family are balanced or rotated across cells, and immediate repetitions of target identity and plane are prohibited.
9. In each trial, the participant begins facing the marked forward direction. A fixation display and spoken target description appear, followed by a provisional 0.5-0.9-second blank interval. All 56 objects then appear, seven on each of eight surrounding vertical planes at 45-degree intervals, and a dedicated `search\_onset` event starts timing. The participant finds one target among distractors and selects by maintaining gaze hover for approximately 1.6 seconds. The 1.6-second dwell is a selection parameter, not a fixation definition. A wrong capture produces neutral feedback and search continues; correct capture ends the trial.
10. A “hint opportunity” is a pre-scheduled time at which one prompt should play; “hint exposure” is a prompt that completed playback. The planned opportunity schedule begins 4 seconds after `search\_onset` and repeats

every 6 seconds until capture. In gaze-contingent blocks, the coarse stage plays a brief spatialized utterance from the target plane when that plane is outside the attended sector; after orientation to the plane, the fine stage maps angular gaze-target distance to discrete colder/warmer/very-close feedback. The policy abstains when gaze is invalid or confidence is below threshold and uses hysteresis/cooldown. In noncontingent blocks, the assistant receives no gaze, hover, coverage, or target-proximity state and plays duration-matched general prompts at the same opportunities. Every opportunity logs evidence, confidence, policy stage/state, utterance/voice identifier, playback onset/outcome, and abstention. **[OPEN DECISION: freeze filters, confidence threshold, plane-entry rule, angular bands, hysteresis, exact approved scripts, maximum trial duration, and any maximum hint count after formative pilot testing and before confirmatory collection.]**

11. Offer a break after every block and any time requested. Stop immediately for concerning symptoms or participant request.
12. Complete the six raw NASA-TLX subscales after each block without pairwise weighting, following Hart and Staveland (1988), and brief guidance/voice manipulation, trust, helpfulness, reliance, comfort, and control checks. Complete demographics, experience, comparative ratings, and open-ended questions once at the end unless collected earlier as specified.
13. Remove the headset, complete a symptom check, explain the study, answer questions, provide compensation under the approved plan if offered, and provide contact information.
14. Verify transfer of coded research logs to approved UCF storage; delete the local source recording, private clone, source sample, generated self-similar clips/history where possible, and device TTS cache under the approved deletion procedure. Record the result. Treat any failed deletion of an identifiable voice asset as an immediate privacy incident: disable/revoke access to the clone, retry and escalate to the provider, notify the PI, assess UCF reporting, and suspend further voice collection while material exposure remains unresolved.

Participants may pause or stop at any time. No procedure involves physical contact other than optional researcher assistance adjusting the headset after permission.

Long-term follow-up: N/A.

Recontacting participants: N/A, except to provide compensation if offered or respond to a participant request.

[OPEN DECISION: confirm.]

15.0 Setting

The study will occur in **[OPEN DECISION: UCF building, lab, and room]**. The room will have a marked seated origin, swivel chair, clear rotational area, institution-managed encrypted researcher workstation, sanitized headset, microphone, and secure network. The release build will instantiate and validate eight vertical virtual search planes around the seated origin at 45-degree intervals. The door will be closed or access controlled during voice recording. A trained researcher will remain present.

The headset uses video-passthrough cameras to render the immediate environment locally. The current application does not save or transmit passthrough frames, room video, webcam video, or face video. Raw eye-camera images are not collected. Any future camera recording or image transfer requires prior IRB modification and updated consent.

Online procedures: N/A, except any approved pre-screening or scheduling form.

International/multi-site locations: N/A.

16.0 Study Timelines

Participant duration: **[OPEN DECISION: proposed 75–90 minutes in one visit.]**

Enrollment period: **[OPEN DECISION.]**

Analysis and study completion: **[OPEN DECISION.]**

No recruitment or research data collection will begin before written UCF approval. The study will close or seek continuing review/status updates as UCF requires.

17.0 Resources Available

Required resources include an eligible UCF PI, trained study staff, approved lab, HTC Vive Focus Vision, marked seated origin, stable swivel chair and clear rotational area, validated eight-plane virtual geometry, institution-managed encrypted staff workstation, compatible secure network, sanitized face interface, private microphone setup, UCF-approved encrypted archival storage, licensed/approved voice provider account, secure secrets management, data-transfer/deletion procedures, and time for consent and breaks.

All team members interacting with participants or identifiable data will be listed in Huron and complete required CITI training before beginning work. Staff will be trained on consent, headset safety, calibration, symptom response, data handling, deviations, adverse events, and vendor deletion.

OPEN DECISION: Confirm whether information-security, privacy, purchasing/contracts, export-control, accessibility, or other ancillary review is required for the voice service and headset data path.

18.0 Potential Benefits to Subjects

Participants should not expect direct benefit. They may find the task interesting or learn generally about mixed-reality research. The research may contribute knowledge about adaptive assistance and voice similarity, but benefit to others is uncertain.

19.0 Risks to Subjects

Physical and sensory risks include temporary nausea, dizziness, disorientation, eyestrain, headache, fatigue, blurred vision, balance difficulty, or discomfort from headset fit. A rare seizure may be triggered in susceptible persons. Mitigations include screening, clear area, stable posture, fit adjustment, practice, frequent breaks, researcher monitoring, immediate stopping, headset removal, seated recovery, and emergency response when needed.

Privacy/confidentiality risks include unauthorized access to contact information, gaze and performance data, or identifiable voice assets. A voice recording or clone may identify the speaker. A synthetic clone can produce speech the participant never recorded, creating an impersonation or misuse concern. Mitigations include explicit authorization, a fixed neutral script, private recording, study IDs, separate linkage, encrypted approved storage, least-privilege access, no public sharing, no Voice Library publication, model-improvement opt-out, restricted prompt set, and documented deletion.

Third-party risk: The approved voice provider will receive source audio and study-authored prompt text and will create/process voice data. Provider systems and policy may not permit the team to guarantee immediate removal from every backup, security, or legal-retention system. ElevenLabs currently states that some user data may be used to improve models unless opted out and that generated voice data may be retained for up to three years after the last interaction, subject to deletion requests and policy exceptions. No participant audio will be uploaded until UCF approves the provider and the final consent accurately states the tested deletion limits.

Psychological/social risks include discomfort, self-consciousness, or an uncanny feeling when hearing a self-similar synthetic voice; frustration with incorrect or delayed hints; and concern about being evaluated. The system is not a diagnostic test, individual scores will not be shared with instructors/employers, participants may pause/stop, and staff will use nonjudgmental language.

Unforeseeable risks are possible because mixed-reality and voice-cloning systems evolve. New material risks or vendor changes will be submitted as a modification before use.

Pregnancy risks: No pregnancy-specific risk is known beyond ordinary headset discomfort; pregnancy is not an exclusion in this draft. UCF must confirm.

Risks to others: A cloned voice could be misused if accessed. The restricted prompt set, access controls, no sharing, and deletion procedure mitigate this risk.

Minimal-risk analysis: The team expects temporary headset symptoms to be comparable in probability and magnitude to ordinary use of a consumer mixed-reality headset with the listed screening and stop controls. The identifiable voice upload, synthetic-voice impersonation potential, and residual provider retention are less routine and are the load-bearing uncertainty in the proposed minimal-risk classification. The team will provide UCF with the final provider controls, deletion test, retention terms, and restricted-prompt workflow so UCF can compare the probability and magnitude of harm with the regulatory ordinary-life/routine-examination baseline. If UCF

determines the study exceeds minimal risk, the protocol and consent will be revised to include the required injury, monitoring, and review provisions before enrollment.

20.0 Compensation for Research-Related Injury (for more than minimal risk studies only—if this is a minimal risk study, check N/A and skip to the Withdrawal of Subjects section)

N/A is proposed because the study is expected to be minimal risk. **[UCF must confirm the risk level and whether specific injury/cost language is required in the protocol and consent.]** Staff will follow emergency procedures and document/report research-related problems as required.

21.0 Withdrawal of Subjects

Participants may decline any nonessential question, request a break, or withdraw at any time without penalty. Because voice recording/cloning and eye tracking are core procedures, a person who declines either before starting cannot participate in the main experiment but will not be penalized.

The researcher may stop participation for concerning symptoms, inability to safely use the headset or rotate while seated, repeated calibration failure, eight-plane geometry/origin validation failure, equipment/vendor failure, protocol noncompliance that creates risk, or study suspension. If compensation is offered, stopping for safety will not reduce compensation already earned under the approved prorating plan.

On withdrawal, no new data will be collected. The participant may request deletion of identifiable source voice assets and the linkage key to the extent they remain under team control. Data already de-identified or included in aggregate analysis may not be retrievable. The consent will accurately state these limits.

22.0 Economic Burden to Subjects (If there are no economic burdens to subjects, check N/A and skip to the Compensation section)

No study fee is proposed. Participants may incur ordinary travel, parking, and time costs. **[OPEN DECISION: confirm whether parking reimbursement or any participant-purchased equipment/data is involved.]**

23.0 Compensation

Are participants compensated? **[OPEN DECISION: yes/no.]**

If yes, specify amount, form (cash/check/gift card/course credit), funding source, payment records, timing, prorating, no-show policy, and withdrawal payment. Compensation will not depend on task performance, correct responses, eye-tracking quality, voice-clone quality, or analyzability. If course credit is offered, an equivalent nonresearch alternative will be available.

24.0 Data Validity

The team will use a frozen build/configuration, nonrepeating deterministic stimulus manifests balanced across sequences, the four-sequence Williams schedule, exact per-condition target-plane quotas, practice trials, calibration/validation checks, session checklist, event logs, and automated file validation. Researcher notes will document deviations and technical failures. Generic/self-similar versions of each semantic prompt will be matched or normalized for wording, duration range, loudness, intelligibility, spatial rendering, and delivery settings; technical quality-control results will be recorded. Provider product/model/version and synthesis settings will be recorded for every generated library. A provider/model or material quality drift will pause collection pending PI review and, when required, an IRB modification.

Before analysis, the PI will approve objective exclusion rules. Pre-assignment eligibility/calibration failures will be distinguished from post-assignment tracking failures. Proposed reason codes include: consent withdrawn; ineligible; calibration failure after specified retries; eight-plane geometry/origin validation failure; post-assignment tracking availability below **[OPEN DECISION threshold]**; missing `search\_onset` or response event; corrupted/incomplete file; incorrect build/configuration/schedule; major interruption; and noncompletion. A wrong selection does not exclude a trial: primary search time continues from `search\_onset` through the eventual correct capture, and wrong selections are analyzed as a secondary count/outcome. The primary assignment-policy analysis will retain assigned participants/trials to the extent an outcome is observable; per-protocol and missing-

data sensitivity analyses will assess tracking-dependent failures. Outlier handling and minimum valid-trial requirements are **[OPEN DECISION]**. Exclusions will not be based on condition results, manipulation-check ratings, or voice similarity and will not affect compensation, if offered.

Participant disposition will report screened, enrolled, started, completed, withdrawn, excluded by reason, and analyzed. Coded raw task, telemetry, and questionnaire files will be retained under the approved policy so exclusions are auditable; identifiable voice assets follow their separate approved deletion schedule.

25.0 Sharing Results with Subjects

Individual performance or eye-tracking results will not be returned because they are research measures, not diagnostic assessments. Participants may request a plain-language aggregate summary after the study by contacting the team. **[OPEN DECISION: whether/how summaries will be offered without retaining an unnecessary contact list.]**

26.0 Data Management and Confidentiality

Identifiable data access/collection:

- Contact/scheduling information: yes, if a participant contacts the team.
- Signed consent and compensation records: yes.
- Voice recording: yes.
- Voice data/private clone and self-similar synthesized speech: yes; potentially identifiable.
- Eye tracking, interaction telemetry, performance, questionnaires, and limited demographics: coded with a study ID.
- Video/face/room recording: no.
- Protected health information or medical records: no.
- Biospecimens: no.

Data flow:

1. Contact/screening and consent records are stored separately from research data.
2. A study ID is assigned. The identity-to-ID key is encrypted and restricted to **[OPEN DECISION: roles]**.
3. The source voice file is recorded on an institution-managed encrypted workstation using only the study ID, uploaded over encrypted TLS to the approved IVC account, and used to create a private clone and fixed study prompts. The temporary folder will not synchronize to consumer/cloud backup.
4. Prepared speech clips are copied to the headset without provider credentials. The headset writes gaze and trial files to `Application.persistentDataPath/GazeData` in participant/run folders. Generated/prepared TTS clips may be cached locally. The headset and workstation are institution-managed, encrypted or protected by an institution-approved equivalent control, access-restricted, and not configured to back up study folders automatically. These device locations are temporary working storage, not the archival repository.
5. After the session, coded logs/questionnaires are transferred to **[OPEN DECISION: approved UCF encrypted storage]**, checked for completeness/checksum, and removed from the headset after verified transfer.
6. Source recording, provider sample/clone/history, self-similar generated clips, and device cache are deleted on the approved schedule, with operator/date/result recorded. A failed deletion triggers immediate access containment, PI escalation, UCF reportability assessment, and a pause on further voice collection while exposure remains unresolved. **[OPEN DECISION: final timing and vendor residual-retention language.]**

Access will be limited by role. Consent/payment records and the linkage key will not be stored with gaze/questionnaire data. Filenames, provider voice names, application logs, and CSV files will not contain names, emails, or initials. Provider credentials will be stored only on the approved staff workflow in an institution-approved secrets mechanism, rotated/revoked on suspected exposure or staff change, and never embedded in the headset release build or study datasets.

Retention: Coded research data, approved protocol/materials, consent records, audit logs, and analysis code will be retained for at least **[OPEN DECISION: UCF-required period; working proposal is five years after study closure or publication, whichever UCF requires]**. The PI will ask UCF whether identifiable voice assets may/should be deleted earlier after derived measures and audit fields are secured. The consent will match the approved schedule.

Data sharing: De-identified aggregate results and possibly a de-identified coded dataset may be published. Raw voice recordings, clone identifiers, generated self-similar clips, identity key, contact information, consent/payment records, and precise raw data that UCF determines could reasonably identify a person will not be publicly shared. **[OPEN DECISION: repository and data-sharing plan.]**

No Certificate of Confidentiality is currently proposed. **[UCF must confirm eligibility/need.]**

27.0 Long-Term Data and/or Specimen Banking (If there are no plans for long-term data and/or specimen banking, check N/A and skip to the Data Analysis Plan section)

N/A. The study does not collect specimens and does not propose a repository for future unspecified research. Retention of coded research records for reproducibility and UCF recordkeeping is described above and is not permission to reuse identifiable voice assets in another study.

28.0 Data Analysis Plan

The primary analysis will model trial-level search time across the 2×2 within-participant conditions with a log-normal mixed-effects model. Fixed effects will include guidance policy, voice identity, their interaction, centered trial index, block position, target azimuth/plane, and initial target visibility. Participant random intercepts and within-participant slopes for both manipulated factors will be included when estimable. Search time runs from explicit `search\_onset` through correct capture, including time spent on preceding wrong captures. The estimand is assignment to the complete guidance policy, not receipt of any individual hint. The guidance main effect and guidance-by-voice interaction on search time are the focal contrasts; Holm's sequential procedure will control multiplicity across the two tests.

Wrong-capture counts will use a negative-binomial generalized mixed model, and first-attempt success will use a logistic mixed model. Head-yaw travel, target-plane latency, plane revisits, and hint-response latency will use distributions selected from blinded pilot diagnostics and frozen before confirmatory analysis. Post-block continuous ratings will use linear mixed models; ordinal models will be used when interval assumptions are untenable. Models will include counterbalancing sequence/order/prior condition and distinguish confirmatory from exploratory analyses. Missingness and exclusions will be summarized by condition. Voice-similarity and perceived-guidance-responsiveness checks support causal interpretation but are not post-assignment exclusion criteria. Because participants can recognize the self-similar voice, voice-condition blinding is not possible. The design supports relative claims about gaze-contingent versus noncontingent policies, not either policy versus no assistance. The analysis will not relabel XRI hover/dwell inspection proxies as validated fixation or saccade measures.

A simulation-based sensitivity and power analysis using pilot variance components must confirm the target of 48 complete datasets and determine the enrolled maximum before preregistration. The plan and code will be timestamped before condition-effect analysis.

29.0 Provisions to Monitor the Data to Ensure the Safety of Subjects (for more than minimal risk studies only—if this is a minimal risk study, check N/A and skip to the Provisions to Protect the Privacy Interests of Subjects section)

N/A is proposed because the study is expected to be minimal risk and no independent data-safety monitoring board is proposed. Session staff will nevertheless monitor symptoms and technical/privacy incidents in real time; the PI will review events, deviations, withdrawals, tracking failures, and deletion failures at **[OPEN DECISION: specified interval]** and pause the study if risk or failure patterns exceed the approved stopping criteria. UCF must confirm whether this monitoring description is sufficient.

30.0 Provisions to Protect the Privacy Interests of Subjects

Consent, screening, and voice recording will occur in a private area. Participants will be addressed respectfully and may skip nonessential questions, take breaks, or stop. Only necessary demographics will be collected. The

headset view and researcher monitor will be positioned to reduce observation by passersby. No individual results will be disclosed to instructors, employers, or other participants.

The research team will explain what the headset tracks and demonstrate neutral audio output before experimental trials without revealing condition labels or directional hypotheses. The source script avoids names and sensitive content. The clone will only synthesize a fixed, approved set of neutral task phrases on the staff-controlled workflow. Provider account access will be limited to designated staff with individual least-privilege accounts, multi-factor authentication when supported, revocable credentials, and an auditable generation/deletion record. Staff will not imitate participants, generate jokes/personal statements, or use the voice outside the session.

31.0 Adverse Events, Unanticipated Problems, and Deviations

For symptoms or immediate risk, the researcher will stop the task, remove the headset, seat the participant, assess whether emergency assistance is needed, and follow UCF/lab emergency procedures. The researcher will document observations and participant-reported symptoms without making a diagnosis, notify the PI promptly, and preserve relevant device/build records.

The PI will assess and report adverse events, unanticipated problems, confidentiality incidents, and serious/continuing noncompliance according to current UCF requirements. Changes will not be implemented before UCF approval except when necessary to eliminate an immediate hazard; emergency changes will be reported promptly.

Protocol modifications requiring prior review include changes to voice provider/model/account/data use, recording duration/script, deletion/retention, camera or audio collection, telemetry fields, eligibility, recruitment, compensation, trial/block schedule, counterbalancing, hint timing/content, outcomes, sample size, sites, or study team.

32.0 Records

The PI will maintain current and superseded approvals/protocols/consents/materials; Huron correspondence; study-team/CITI/delegation records; screening/enrollment and consent logs; build/configuration and session records; counterbalancing assignments; data-transfer and deletion logs; compensation records if applicable; deviations/events and corrective actions; data dictionaries; coded raw task/telemetry/questionnaire data; analysis code/outputs; publications; and closure records for the approved retention period. Identifiable source voice recordings, provider samples/clones/history, and self-similar clips follow the separately approved early-deletion schedule and are not included in the general “raw data” retention statement.

33.0 Rationale for the Voice-Cloning Factor and Provider Path

The self-similar voice factor is a core scientific aim of the 2 × 2 design because the study tests whether voice similarity changes the effect of gaze-contingent assistance, not merely whether gaze contingency works with a generic voice. The final hypothesis hierarchy and power analysis must reflect that interaction; if it remains only a low-priority exploratory outcome, the PI will reconsider whether identifiable voice cloning is proportionate to the added risk.

OPEN DECISION: Before submission, compare an institution-hosted/on-device voice-conversion or cloning workflow with the proposed approved cloud IVC workflow. Document voice quality, latency/session feasibility, security review, deletion control, contracting, and reproducibility. Use the lower-risk path that can validly deliver the manipulation. If no acceptable voice path is approved, remove the voice factor through a protocol revision rather than collecting identifiable voice data under an unapproved compromise.

34.0 Multi-Site Research

N/A. This is currently a single-site UCF study. Adding a site or relying institution requires a modification and reliance/IRB determination before that site begins work.

35.0 References and External Services

- UCF Office of Research, IRB and current Huron guidance: <https://www.research.ucf.edu/compliance/irb/>
- Federal Policy for the Protection of Human Subjects, 45 CFR 46: <https://www.ecfr.gov/current/title-45/subtitle-A/subchapter-A/part-46>

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- ElevenLabs voice-cloning concepts: <https://elevenlabs.io/docs/eleven-api/concepts/voice-cloning>
- ElevenLabs privacy policy: <https://elevenlabs.io/privacy-policy>

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Adult Consent to Take Part in Research

Protocol proposal: This draft describes the intended participant experience. It must not be used until all open decisions are resolved and UCF approves the study and consent form.

Study title: **Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search**

Principal Investigator: **[OPEN DECISION: name and UCF department]**

Version/date: **[OPEN DECISION]**

Key Information

You are invited to take part in a research study about spoken assistance during a mixed-reality visual-search task. If you join, you will come to a UCF lab for about **[OPEN DECISION: 75–90 minutes]**. While seated in a swivel chair, you will wear an HTC Vive Focus Vision headset and turn to search for virtual colored shapes arranged around you. You will complete practice followed by four 16-trial blocks (64 experimental trials), receive spoken hints, answer questionnaires, and provide about 1–2 minutes of your own speech. A third-party speech service will use that recording to make a private synthetic voice that resembles you and can say study phrases you did not record.

The main risks are temporary nausea, dizziness, eyestrain, headache, fatigue, or discomfort while using the headset; frustration or discomfort with the task or self-similar voice; and loss of privacy involving your gaze, performance, or identifiable voice information. You may pause or stop at any time. You should not expect a direct benefit. Participation is voluntary.

Why am I being invited to take part in a research study?

You are being invited because you are an adult who may be able to safely use a mixed-reality headset, understand the study in English, and provide your own voice recording. Screening questions will confirm eligibility.

Why is this research being done?

The researchers want to learn whether spoken hints based on where a person looks affect visual-search performance and workload, and whether a voice that resembles the participant changes trust, comfort, or experience. This is not a medical or diagnostic test.

How long will the research last and what will I need to do?

Your participation will involve one lab visit lasting about **[OPEN DECISION: proposed 75–90 minutes]**. You will review this form, complete safety screening, record a short voice sample, wear and calibrate a mixed-reality headset, complete up to eight practice trials and 64 experimental search trials in four blocks, take breaks, answer questionnaires, and receive a study explanation.

Is there any way being in this study could be bad for me?

The headset may cause nausea, dizziness, disorientation, eyestrain, headache, fatigue, blurred vision, balance difficulty, or pressure/discomfort. Hearing a synthetic voice that resembles you may feel unusual, uncanny, or uncomfortable. Spoken hints may be delayed, inaccurate, or frustrating.

Your voice is identifiable. A synthetic voice can produce words you never recorded. If the recording, private voice clone, or generated clips were accessed or misused, someone might recognize or imitate your voice. The study uses a restricted script, coded filenames, access controls, private settings, no public voice sharing, and a deletion

procedure to lower this risk, but complete confidentiality and deletion from every provider backup cannot be guaranteed.

Will being in this study help me in any way?

You should not expect a direct benefit. The study may help researchers understand how to design mixed-reality assistance, but that benefit is uncertain.

What happens if I do not want to be in this research?

You do not have to participate. Saying no will not affect your relationship with UCF, your grades, employment, services, or benefits. **[If course credit is used: add the approved equivalent alternative assignment.]**

What should I know about a research study?

Taking part is voluntary. You may ask questions before deciding. You may skip any nonessential question, request a break, or stop later without penalty. The researchers may stop your participation for safety, ineligibility, repeated calibration failure, or equipment failure.

Who can I talk to?

For questions, concerns, complaints, research-related problems, or to request withdrawal/deletion where possible, contact:

- Principal Investigator: **[OPEN DECISION: name, phone, UCF email]**
- Faculty Advisor: **[OPEN DECISION: name, phone/email, if applicable]**

For questions about your rights as a research participant or to speak with someone independent of the research team, contact the UCF Institutional Review Board at 407-823-2901 or irb@ucf.edu. **[Confirm current contact wording against the approved HRP-502 template before submission.]**

How many people will be studied?

The researchers plan to obtain complete study data from 48 people. Up to **[OPEN DECISION: enrolled maximum after accounting for screening, withdrawal, and technical attrition]** people may be enrolled at UCF.

What happens if I say yes, I want to be in this research?

After you sign this form, a researcher will:

1. Assign a study ID that does not include your name.
2. Ask short questions about headset safety and eligibility.
3. Ask you to read a neutral script for about 1–2 minutes in a private room. The recording will be made on an institution-managed encrypted research computer.
4. Upload the recording over an encrypted connection to **[OPEN DECISION: approved provider; proposed ElevenLabs Instant Voice Cloning]** to create a private synthetic voice. On the protected research computer, the team will generate only the fixed study instructions and hints for your session and then copy those audio clips to the headset. The headset will not contain the provider password/key or send your recording to the provider. The voice will not be published or shared to a public voice library.
5. Seat you in a swivel chair at a marked position, fit and calibrate the HTC Vive Focus Vision headset, and check that you can comfortably turn to view the search area around you. The headset uses cameras to show your surroundings and eye tracking to estimate where you look. The study application does not save room video, face video, webcam video, or raw eye-camera images.
6. Ask you to complete up to eight practice trials, then four 16-trial experimental blocks. In each experimental trial, you will turn while remaining seated to find one target among 56 virtual colored shapes arranged across eight virtual areas around you and hold your gaze on your selection. In some blocks, hints will use gaze-related task information to provide coarse directional guidance followed by closer/farther feedback; in other blocks, general hints will not use that information. The same hint wording will be spoken either in a generic voice or in the self-similar synthetic voice.

7. Offer breaks and ask about symptoms, workload, the hints, and the voices.
8. Remove the headset, explain the study, and provide compensation under the approved plan if compensation is offered.

The system records coded eye-gaze rays and availability, eye openness when available, what virtual object is under the gaze pointer, dwell progress, trial events, hint timing, task performance, application state, and questionnaire responses. These measures are research interaction data and are not used to diagnose you.

What are my responsibilities if I take part in this research?

Tell the researcher immediately if you feel dizzy, nauseated, disoriented, have eye pain, headache, blurred vision, neck or back discomfort, anxiety, or any other concern. Follow the clear-area instructions, remain seated in the approved task position, rotate only as instructed, do not share personal information in the voice script, and answer safety questions honestly.

What happens if I say yes, but I change my mind later?

You may stop at any time. Tell the researcher and the task will stop. No new information will be collected. You may ask the team to delete your identifiable source voice recording, private clone, and identity link if they still exist under the team's control. Information already stripped of the identity link or included in combined results may not be retrievable. A provider may keep limited information in backups, security logs, or as required by policy/law even after the research team submits a deletion request.

If compensation is offered, stopping will not reduce compensation already earned under the approved payment plan. **[OPEN DECISION: insert exact prorating.]**

Is there any way being in this study could be bad for me? (Detailed Risks)

Mixed-reality risks: temporary motion sickness, dizziness, disorientation, eyestrain, headache, fatigue, blurred vision, balance problems, headset pressure, or neck/back discomfort from turning while seated. A seizure is rare but may occur in susceptible people. The chair and surrounding area will be checked, a researcher will remain present, breaks will be offered between blocks, and you may remove the headset or take a break at any time.

Voice risks: your recording and synthetic voice may identify you. A synthetic voice is not just a replay; it can produce approved study sentences you never said. Accidental disclosure could create embarrassment, impersonation, or other privacy harms. The researchers will use a coded name, fixed phrases, private account settings, limited access, and documented deletion. They will not publicly share or reuse your voice.

Third-party processing: the approved speech provider will receive your audio and the text of fixed study prompts. If ElevenLabs is approved, the research account will turn off the setting that allows new data to improve models. ElevenLabs states that voice data may otherwise be retained under its privacy policy, including for limited operational, security, backup, or legal purposes. **[OPEN DECISION: replace with the final UCF-approved provider and tested retention/deletion statement.]**

Privacy risks: coded gaze, performance, questionnaire, consent, contact, or payment records could be disclosed. The researchers will separate identity records from coded study data and use UCF-approved storage, but confidentiality cannot be guaranteed.

Emotional risks: the self-similar voice or task may feel uncomfortable, uncanny, frustrating, or evaluative. You may pause or stop and may decline nonessential questions.

Unforeseeable risks may exist because voice-cloning and mixed-reality technologies change. Material changes will not be used with participants until reviewed by UCF.

What happens to the information collected for the research?

Your name and contact information will be kept separate from the study files. The study files will use a study ID. Coded data are not anonymous while the identity key exists, and voice recordings/clones may remain identifiable even when labeled only with a code.

The protected research computer temporarily stores your coded voice recording and prepared speech files, and the headset temporarily stores coded gaze/task logs and prepared speech files. Automatic consumer-cloud backup will be disabled for these temporary study folders. After the session, coded research data will be

transferred to **[OPEN DECISION: approved UCF encrypted storage]** and removed from the devices after the transfer is checked. The source voice recording, private clone, provider samples/history, and generated self-similar clips will be deleted on **[OPEN DECISION: final approved schedule]**, and the deletion result will be recorded. If deletion fails, the team will restrict access, notify the Principal Investigator, contact the provider, assess whether UCF must be notified, and pause further voice collection while the problem remains unresolved.

The research team, UCF IRB/oversight offices, and others authorized by law or UCF may inspect study records. **[OPEN DECISION: add sponsor/provider/monitor access if applicable.]**

Coded research data, consent/approval records, and analysis files will be kept for at least **[OPEN DECISION: UCF-required period; proposed at least five years after study closure or publication]**. De-identified combined results may be published. Raw voice recordings, private voice clones, generated self-similar voice clips, names, contact information, signed consent, payment records, and the identity key will not be publicly shared.

Absolute confidentiality cannot be promised. There is a small risk of unauthorized access despite safeguards.

Can I be removed from the research without my OK?

The researchers may stop your participation if continuing appears unsafe, you do not meet eligibility requirements, eye tracking or the eight-area search environment cannot be set up after the approved retry process, equipment/provider service fails, the study is suspended, or you do not follow safety instructions. The reason will be explained when appropriate. If compensation is offered, payment will follow the approved prorating plan and will not depend on usable data.

What else do I need to know?

Compensation: **[OPEN DECISION: amount, form, timing, prorating, and payment-record requirements.]** You will not be paid based on speed, accuracy, eye-tracking quality, voice quality, or whether your data are analyzed.

Costs: There is no cost to you other than travel/time. **[OPEN DECISION: confirm.]**

Research-related injury: **[OPEN DECISION: insert UCF-approved injury/cost language from HRP-502 if required.]**

New findings: If the researchers learn information that may affect your willingness to continue, they will tell you as required.

Conflicts of interest: The investigator reports no financial interest. **[PI/UCF must confirm the Huron conflict-of-interest disclosure before submission.]**

Voice-Cloning Authorization

By initialing below, I confirm that:

- the recording is my own voice;
- I authorize the approved research team to upload it to the approved speech provider and create a private synthetic voice for this study;
- I understand the synthetic voice can say the fixed study phrases even though I did not record those exact phrases;
- I understand the voice will not be published, shared to a voice library, reused for another project, or used for unrestricted/free-form speech;
- I understand the privacy and deletion limits described above.

Participant initials: _____

Voice cloning is required for this experiment. If you do not authorize it, you cannot take part, but there is no penalty.

Consent and Signature

I have read this form or had it read to me. My questions have been answered. I voluntarily agree to take part and will receive a copy of this form.

Participant name: _____

Participant signature: _____ Date: _____ Time: _____

Person obtaining consent: _____

Signature: _____ Date: _____ Time: _____

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Recruitment Materials

Protocol proposal: These are proposed recruitment versions. Use only the channels and final text approved by UCF.

Study: Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

Use only the versions and channels approved by UCF. Replace every open decision before use.

1.0 Recruitment Email

Subject: Participate in a UCF mixed-reality visual-search study

Researchers at the University of Central Florida are inviting adults to take part in a laboratory study about spoken assistance during a mixed-reality visual-search task.

If you participate, you will remain seated in a swivel chair while wearing an HTC Vive Focus Vision headset, turn to look for virtual colored shapes arranged around you, receive spoken hints, answer questionnaires, and record about 1–2 minutes of your own speech. The task includes practice and four 16-trial blocks (64 experimental trials). An approved third-party speech service will use the recording to create a private synthetic voice that resembles you for study prompts. The study also records eye-gaze interaction data and task performance. It does not save room video, face video, webcam video, or raw eye-camera images.

The visit will take about [OPEN DECISION: duration] at [OPEN DECISION: UCF lab/location]. Compensation is [OPEN DECISION: amount/form or state no compensation].

You may be eligible if you are 18 or older, can provide informed consent in English, can safely use a mixed-reality headset, and are willing to provide and authorize use of your own short voice recording. Additional safety screening applies.

Participation is voluntary. Choosing not to participate will not affect your grades, employment, UCF services, or relationship with UCF.

To learn more or complete the eligibility screening, contact [OPEN DECISION: study email/phone or approved screening link].

UCF IRB study number: [add after assignment/approval]

Principal Investigator: [OPEN DECISION]

2.0 Flyer or Research-Portal Listing

Mixed-Reality Visual Search + Your Synthetic Voice

Take part in a UCF laboratory research study about gaze-based spoken assistance.

- One [OPEN DECISION: duration] lab visit
- Wear an HTC Vive Focus Vision headset
- Remain seated and turn in a swivel chair to search around you
- Complete practice and four 16-trial search blocks
- Eye-gaze interaction and task data recorded
- Record 1–2 minutes of your own speech
- Hear private study prompts in a synthetic voice that resembles you
- Answer short questionnaires
- Compensation: [OPEN DECISION]

You may qualify if you are age 18 or older, can consent in English, can safely use a mixed-reality headset and turn while seated, and agree to the short voice recording and private voice-cloning procedure.

Interested? **[OPEN DECISION: approved contact/link]**

Participation is voluntary. UCF IRB study number: **[after approval]**.

3.0 Verbal Referral Script

"A UCF research team is studying spoken assistance in a mixed-reality visual-search task. The study includes eye tracking and a short recording used to make a private synthetic voice that resembles the participant. If you are interested, I can give you the approved study contact or link. I cannot tell you whether you qualify, and there is no obligation to contact the team."

4.0 Recruitment Controls

- Do not recruit or schedule research visits before UCF approval.
- Do not describe participation as risk-free, therapeutic, diagnostic, or guaranteed to benefit the participant.
- Do not hide the eye tracking or voice cloning in short-form ads.
- Do not recruit people under the investigator's direct grading, employment, or supervisory authority unless UCF approves safeguards.
- Do not collect eligibility answers by ordinary email; use the approved screening process.

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Eligibility and Safety Screening

Protocol proposal: This is a proposed screening workflow. The PI and UCF must approve the consent point, eligibility rules, safety thresholds, and record-retention procedure before use.

Study: Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

Researcher instructions: Conduct only after the approved screening consent/consent point specified by UCF. Ask in private. Record the study ID and eligibility result, not diagnostic detail. Do not retain reasons longer than the approved screening-record schedule. "Yes" to a safety item does not authorize the researcher to diagnose the person.

Study ID: _____ Date/time: _____ Screener: _____

1.0 Basic Eligibility

1. Are you 18 years of age or older? ☐ Yes ☐ No
 2. Are you able to read, understand, and provide informed consent in English? ☐ Yes ☐ No
 3. Are you able to understand spoken English instructions and prompts? ☐ Yes ☐ No
 4. Are you willing to provide about 1–2 minutes of your own speech and authorize a private synthetic voice for this study? ☐ Yes ☐ No
 5. Have you previously taken part in this same experiment or learned the target/order schedule? ☐ Yes ☐ No
- Proposed eligibility rule: Questions 1–4 must be Yes; question 5 must be No. **[OPEN DECISION: PI/UCF confirm.]**

2.0 Headset and Visual Safety

6. Has a health professional told you to avoid virtual/mixed-reality headsets, flashing displays, or stereoscopic displays? ☐ Yes ☐ No
 7. Do you have a history of photosensitive seizure or epilepsy? ☐ Yes ☐ No
 8. Right now, are you experiencing severe dizziness, nausea, migraine/headache, eye pain/infection, blurred vision beyond your usual correction, or another symptom that would make headset use unsafe or uncomfortable? ☐ Yes ☐ No
 9. Do you have any concern about safely wearing the headset and repeatedly turning in a swivel chair while remaining seated today, including a current neck, back, balance, or vestibular concern? ☐ Yes ☐ No
 10. Do you use glasses or contact lenses? ☐ No ☐ Glasses ☐ Contact lenses
- If yes, can your ordinary correction be used comfortably with the headset? ☐ Yes ☐ No ☐ Not yet tested
11. Do you use a hearing aid or have difficulty hearing ordinary spoken instructions? ☐ No ☐ Yes
- If yes, can the study audio be adjusted or presented in an approved accessible way? ☐ Yes ☐ No ☐ Not yet tested
12. Can you distinguish the study's approved color set during the standardized headset color check, with your ordinary correction if used? ☐ Yes ☐ No ☐ Not yet tested

Proposed safety rule: A Yes to 6, 7, 8, or 9 requires stopping eligibility and PI review under the approved rule; do not ask for unnecessary medical details. Questions 10–12 require a standardized fit, audio, or color check. Accessibility needs should be evaluated consistently and should not be treated as exclusions unless the task cannot be completed safely or the manipulation cannot be validly delivered. **[OPEN DECISION: finalize.]**

3.0 Voice Recording Safety and Privacy Confirmation

13. Is the voice you will record your own voice? ☐ Yes ☐ No
14. Do you understand that the synthetic voice can say approved study phrases that you did not record word-for-word? ☐ Yes ☐ No
15. Do you understand that the recording will be sent to the approved third-party speech provider described in the consent form? ☐ Yes ☐ No
16. Do you understand that the voice is required for this experiment and that you may choose not to participate without penalty? ☐ Yes ☐ No

Questions 13–16 must be Yes.

4.0 Pre-Session Symptom Rating

Rate each item now: 0 = none, 1 = slight, 2 = moderate, 3 = severe.

Symptom	0	1	2	3
General discomfort	☐	☐	☐	☐
Fatigue	☐	☐	☐	☐
Headache	☐	☐	☐	☐
Eyestrain	☐	☐	☐	☐
Difficulty focusing	☐	☐	☐	☐
Nausea	☐	☐	☐	☐
Dizziness/ disorientation	☐	☐	☐	☐

[OPEN DECISION: define a pre-session postponement/stop threshold and whether the adapted symptom items require the full validated SSQ instead.]

5.0 Researcher Determination

- ☐ Eligible to continue
- ☐ Ineligible—record only approved reason code: _____
- ☐ Postpone/reschedule under approved rule
- ☐ PI consultation required

Researcher signature/initials: _____

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Participant Questionnaires

Protocol proposal: These instruments and administration times are proposed. Finalize the open instrument decisions and obtain UCF approval before participant use.

Study: Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

Use study ID only. Do not collect names in these instruments. Participants may skip any nonessential question.

1.0 Background and Prior Experience (once per participant)

1. Age in years: _____ **[OPEN DECISION: exact age versus age bands.]**
2. Gender identity (select all that apply):
☐ Woman ☐ Man ☐ Nonbinary ☐ Another identity: _____ ☐ Prefer not to answer
3. Do you use glasses or contact lenses?
☐ Neither ☐ Glasses ☐ Contact lenses ☐ Both ☐ Prefer not to answer
4. How often have you used virtual- or mixed-reality headsets?
☐ Never ☐ Once or twice ☐ Several times ☐ Monthly ☐ Weekly or more
5. Have you used eye-tracking technology before?
☐ No ☐ Yes ☐ Unsure
6. Have you used a synthetic or cloned version of your own voice before?
☐ No ☐ Yes ☐ Unsure
7. How familiar are you with AI voice assistants?
 1 Not at all familiar — 7 Extremely familiar: _____

2.0 Raw NASA Task Load Index (after each block)

Mark each scale from 0 (low) to 100 (high). This packet uses the six raw NASA-TLX subscales without pairwise weighting (Hart & Staveland, 1988; [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)).

Dimension	Prompt	Rating 0–100
Mental demand	How mentally demanding was this block?	_____
Physical demand	How physically demanding was this block?	_____
Temporal demand	How hurried or rushed did the pace feel?	_____
Performance	How successful were you in accomplishing what you were asked to do? 0 = perfect, 100 = failure.	_____
Effort	How hard did you have to work to accomplish your level of performance?	_____

Dimension	Prompt	Rating 0–100
Frustration	How insecure, discouraged, irritated, stressed, and annoyed were you?	_____

Block number: _____ Assistance condition code: _____ Voice condition code: _____

These condition-code fields are completed automatically/by the researcher and are not shown as manipulation labels if doing so would bias responses.

3.0 Post-Block Assistance and Voice Check

Rate each statement from 1 = strongly disagree to 7 = strongly agree.

1. The hints seemed responsive to what I was looking at. 1 2 3 4 5 6 7
2. The hints arrived at useful times. 1 2 3 4 5 6 7
3. The hints were helpful. 1 2 3 4 5 6 7
4. The hints were distracting. 1 2 3 4 5 6 7
5. The speaking voice sounded like me. 1 2 3 4 5 6 7
6. The speaking voice felt familiar. 1 2 3 4 5 6 7
7. I trusted the spoken guidance. 1 2 3 4 5 6 7
8. I felt comfortable hearing this voice. 1 2 3 4 5 6 7
9. The voice felt uncanny or unsettling. 1 2 3 4 5 6 7
10. I felt in control of the task. 1 2 3 4 5 6 7
11. The speaking voice sounded natural. 1 2 3 4 5 6 7
12. The hints helped me identify the correct general direction. 1 2 3 4 5 6 7
13. The closer/farther feedback helped me narrow down the target location. 1 2 3 4 5 6 7
14. I relied on the spoken hints when deciding where to search next. 1 2 3 4 5 6 7

Items 1–14 are study-created manipulation, experience, and mechanism checks unless a cited validated scale is adopted before submission; they must not be reported as a single validated score without supporting evidence.

Technical check:

- Did you hear all prompts clearly? ☐ Yes ☐ No ☐ Unsure
- Did any prompt repeat, cut off, arrive late, or use the wrong voice? ☐ No ☐ Yes: _____
- Did you take an unplanned break or experience an interruption? ☐ No ☐ Yes: _____

4.0 Post-Session Comparison

1. Which type of assistance did you prefer?
☐ More gaze-responsive ☐ Less gaze-responsive ☐ No preference ☐ Unsure
2. Which voice did you prefer?
☐ Generic voice ☐ Voice resembling me ☐ No preference ☐ Unsure
3. Which combination felt most helpful? _____
4. Which combination felt most comfortable? _____
5. Did you believe the self-similar voice resembled your voice? Why or why not? _____
6. When the assistant gave directional guidance and then closer/farther feedback, which stage was more useful, and why? _____
7. Did any hint reveal information you had not expected the system to infer from your gaze?

8. Did hearing your self-similar voice create any concern about misuse or about how the voice was created?

9. What would you change about the hints, voice, or task? _____

5.0 Post-Session Symptom Rating

Rate each item now: 0 = none, 1 = slight, 2 = moderate, 3 = severe.

Symptom	0	1	2	3
General discomfort	☐	☐	☐	☐
Fatigue	☐	☐	☐	☐
Headache	☐	☐	☐	☐
Eyestrain	☐	☐	☐	☐
Difficulty focusing	☐	☐	☐	☐
Nausea	☐	☐	☐	☐
Dizziness/ disorientation	☐	☐	☐	☐

Did you stop or pause because of symptoms? ☐ No ☐ Yes

Do you feel ready to leave the lab? ☐ Yes ☐ No—researcher follow-up required

6.0 Instrument Decisions Before Submission

- **OPEN DECISION:** Confirm whether raw NASA-TLX is the chosen workload measure and preserve its standard wording/scoring.
- **OPEN DECISION:** Confirm whether to use the full Simulator Sickness Questionnaire rather than the concise symptom check.
- **OPEN DECISION:** Identify which experience/trust/manipulation items are validated scales versus study-created items; label and analyze them accordingly.
- **OPEN DECISION:** Finalize administration timing, randomization, required/optional items, scoring, and handling of skipped responses.

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Standardized Voice Recording Script

Protocol proposal: This script and upload workflow are proposed. Use only after UCF approves the provider, data path, exact script, and deletion procedure.

Study: Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

1.0 Researcher Setup

- Confirm signed consent and voice-cloning authorization before recording.
- Use a private, quiet room, the approved microphone/settings, and the institution-managed encrypted voice workstation.
- Name the file only with study ID, session/run, and recording version; never use a name or email.
- Confirm that the approved provider account has model-improvement sharing disabled and that the private/no-sharing setting is active.
- Confirm that temporary folders do not synchronize to consumer/cloud backup and that the provider credential is not present on the headset.
- Ask the participant to speak naturally at a consistent volume. Do not coach accent, identity, emotion, or impersonation.
- Record only the approved text. Stop if the participant includes a name or other personal information; delete that take and restart.
- Target approximately 1–2 minutes of clear single-speaker audio, consistent with the proposed IVC path. Upload over encrypted TLS, generate only the approved fixed prompt manifest on the staff workstation, and copy prepared clips—not credentials—to the headset.

2.0 Participant Introduction

"I will ask you to read a short neutral passage in your normal speaking voice. Please do not say your name or add personal details. The approved speech service will use this sample to make a private synthetic voice for this study. The synthetic voice can say the study's fixed phrases even though you did not record those exact phrases. You may stop before or during recording. If you stop, the recording will be deleted and you will not continue in the main experiment, with no penalty."

Confirm: "Do you want to continue with the recording?" ☐ Yes ☐ No

3.0 Recording Passage

Today I am taking part in a visual search activity. While seated, I will turn to look at several areas arranged around me and compare objects by color and shape. Some objects may look similar, so I will take my time and check each area carefully.

The assistant may offer a short hint. A hint can tell me that I am closer, farther away, or looking in a useful direction. I can listen to the hint, keep searching, and make my own choice. If I need a break, I can pause the activity.

Clear speech can change with pace, emphasis, and punctuation. I can say, "Look this way," "colder," "warmer," or "very close." I can ask, "Is the blue shape beside the red one?" I can count: one, two, three, four, five, six, seven, eight, nine, ten.

The quick brown fox moves past the quiet garden. Bright yellow stars appear above purple cubes. Round spheres, narrow cylinders, and small pyramids fill the display. Each sentence gives the system a different pattern of sounds while keeping the content neutral.

I understand that this recording is for the approved research study. I will not say my name, address, phone number, or any other personal information in this recording.

4.0 Quality and Upload Record

Study ID: _____ Recording version/take: _____

Start/end time: _____ Duration: _____

Approved microphone/settings: _____

Background noise/reverb acceptable? ☐ Yes ☐ No

Contains only one speaker and no identifiers? ☐ Yes ☐ No

Participant confirmed continuation immediately before recording? ☐ Yes ☐ No

File checksum (if used): _____

Provider/product/model/version/settings: _____ Account setting evidence ID:

Private voice alias: _____ Upload result: _____

Fixed prompt manifest/generation log: _____ Copied to headset: ☐ Yes ☐ No

Headset provider credential/API call absent: ☐ Yes ☐ No

If quality fails, do not retain failed takes beyond the approved immediate cleanup step. **[OPEN DECISION: maximum retry count and failure/fallback rule.]**

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Researcher Session, Deviation, and Event Record

Protocol proposal: This checklist records the intended approved workflow. It does not establish that the workflow is implemented or authorized for participant use.

Study: **Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search**

This internal record uses study ID only. Do not record diagnoses or unnecessary identifying details. Store signed consent, contact/payment records, and the identity key separately.

Study ID: _____ Session ID: _____ Date: _____

Researcher: _____ Location: _____

1.0 Pre-Session Controls

- ☐ Current UCF approval confirmed; study not expired/suspended.
- ☐ Researcher listed on study and CITI/training current.
- ☐ Correct approved consent/material versions available.
- ☐ Private room, marked seated origin, swivel chair, clear rotational area, and emergency contacts ready.
- ☐ Headset/contact surfaces sanitized; equipment safety check passed.
- ☐ Approved release build/configuration loaded.
- ☐ Build commit/hash: _____ App version: _____
- ☐ Deterministic stimulus manifest/version: _____
- ☐ Counterbalancing schedule version: _____ Assigned Williams sequence: ☐ A-B-D-C ☐ B-C-A-D ☐ C-D-B-A ☐ D-A-C-B
- ☐ Allocation was revealed from the pre-generated randomized schedule only after study ID assignment; assigned sequence was not selected by staff.
- ☐ Approved voice provider/account; model-improvement sharing disabled; private/no-library-sharing controls confirmed.
- ☐ Institution-managed encrypted voice workstation ready; automatic cloud backup/sync disabled for temporary study folders.
- ☐ Provider access uses designated staff, least privilege, MFA when supported, and revocable credentials.
- ☐ Provider credential is restricted to the staff workflow and is not embedded in the headset build/logs.
- ☐ Marked origin and eight-plane 360-degree search geometry validated at 45-degree intervals; all 56 object locations are visible from the seated rotation area.

2.0 Consent and Eligibility

- Consent version/date: _____
- Consent obtained by: _____ Date/time: _____
- ☐ Participant received/offered a copy.
- ☐ Voice-cloning authorization initialed/signed.
- ☐ Comprehension confirmed.
- Screening result: ☐ Eligible ☐ Ineligible ☐ Postponed ☐ PI review
- Approved screening reason code if not eligible: _____
- Pre-session symptom stop threshold triggered? ☐ No ☐ Yes—do not proceed

3.0 Voice Procedure

- Recording script/version: _____ Take: _____ Duration: _____
- ☐ No name or direct identifier in recording/file/provider alias.
- Provider/product/model/version/settings: _____
- Voice alias: _____ Voice ID stored only in restricted deletion log: ☐ Yes
- Upload/create result: ☐ Success ☐ Failed ☐ Not attempted
- Fixed study prompts generated on staff workstation and spot-checked: ☐ Yes ☐ No
- Prompt-library manifest/generation log: _____ Copied to headset: ☐ Yes ☐ No
- Headset provider/API network call disabled/absent: ☐ Yes ☐ No
- Generic voice version/model: _____
- Voice-clone or synthesis issue/fallback: _____

4.0 Headset, Calibration, and Practice

- Headset asset ID: _____ Eye-tracking provider/version: _____
- Eight-plane geometry/origin check: ☐ Pass ☐ Fail Result/retry: _____
- Fit/vision correction acceptable: ☐ Yes ☐ No
- Eye calibration result: ☐ Pass ☐ Fail Retries: _____ Quality/validity output: _____
- Multi-direction gaze validation result: ☐ Pass ☐ Fail Directions/targets checked: _____
- Audio check: ☐ Pass ☐ Fail Volume/configuration: _____
- Practice trials attempted: _____ (maximum 8) Consecutive correct: _____ (criterion 3) Criterion met: ☐ Yes ☐ No
- Dwell selection understood: ☐ Yes ☐ No
- Participant chose to continue after practice: ☐ Yes ☐ No

5.0 Experimental Blocks

Block	Assistance	Voice	Planned trials	Valid trials	Start/end	Break offered/taken	Notes/reason codes
1			16				
2			16				
3			16				
4			16				

- ☐ Raw NASA-TLX completed after each completed block.
- ☐ Manipulation/comfort checks completed after each completed block.
- ☐ End-of-session instruments completed or skips recorded without pressure.
- ☐ Each condition manifest places the target exactly twice on each of the eight planes.
- ☐ Search-onset, scheduled hint opportunity (first at 4 seconds; then every 6 seconds), policy evidence/state, playback outcome, selection, and trial-end events logged.
- ☐ Gaze-contingent blocks used coarse directional then fine proximity guidance; noncontingent blocks did not receive gaze, hover, coverage, or target-proximity state.

6.0 Data Transfer and Cleanup

- Coded data destination: _____ Transfer time/operator: _____
- File manifest/checksum validation: ☐ Pass ☐ Fail Record: _____
- Headset research logs removed after verified transfer: ☐ Yes ☐ No ☐ N/A

- Local source recording deleted: ☐ Yes ☐ No Date/time/result: _____
- Provider source sample deleted: ☐ Yes ☐ No ☐ Not supported Date/time/result: _____
- Provider private voice deleted: ☐ Yes ☐ No ☐ Not supported Date/time/result: _____
- Provider generated history/clips deleted: ☐ Yes ☐ No ☐ Not supported Date/time/result: _____
- Device TTS cache/self-similar clips deleted: ☐ Yes ☐ No Date/time/result: _____
- Identity/linkage record updated under approved plan: ☐ Yes ☐ No
- Any deletion failure: access disabled/revoked ☐ N/A ☐ Yes PI/provider notified ☐ N/A ☐ Yes
- UCF reportability assessed and further voice collection paused while unresolved: ☐ N/A ☐ Yes

7.0 Session Disposition and Data-Quality Reason Codes

Disposition: ☐ Completed ☐ Participant withdrew ☐ Researcher stopped ☐ Technical stop ☐ Ineligible after start

Compensation amount/form issued or owed: _____ Payment record location: _____

Pre-specified reason codes (select all that apply; do not decide based on condition outcome):

- ☐ None known at session end
- ☐ Consent withdrawn
- ☐ Calibration failure after approved retries
- ☐ Eight-plane geometry/origin validation failure
- ☐ Tracking availability below approved threshold
- ☐ Missing/corrupt search-onset or response event
- ☐ Corrupted/incomplete file
- ☐ Wrong build/configuration/schedule
- ☐ Voice condition generation/playback failure
- ☐ Major interruption/protocol deviation
- ☐ Participant did not complete
- ☐ Other approved code: _____

Final analysis exclusion is determined under the approved analysis plan, not by the session researcher.

8.0 Symptoms or Adverse Event

Did the participant report or display a symptom/incident? ☐ No ☐ Yes

If yes:

- Date/time and block/trial: _____
- Observable facts and participant's words (no diagnosis): _____
- Task stopped and headset removed: ☐ Yes ☐ No
- Seated/rested and monitored: ☐ Yes ☐ No
- Emergency/medical assistance offered or activated: _____
- Outcome before departure/transfer of care: _____
- PI notified date/time: _____
- UCF reportability assessment and Huron report/reference: _____
- Corrective/protective action: _____

9.0 Protocol Deviation or Confidentiality Incident

Deviation/incident occurred? ☐ No ☐ Yes

- Date/time: _____
- Approved requirement versus what occurred: _____
- Immediate safety/privacy effect: _____

- Data affected: _____
- Immediate containment/correction: _____
- PI notified: _____
- UCF report/modification reference if required: _____
- Prevention/follow-up: _____

Researcher signature/initials: _____ Date/time completed: _____

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Post-Participation Information

Protocol proposal: Provide this information only in the final UCF-approved form and at the approved point in the study procedure.

Study: Follow My Voice: Two-Step Gaze-Contingent Guidance with Self-Similar Voices for 360-Degree Mixed-Reality Search

Thank you for taking part. This study compares a generic voice with a private synthetic voice derived from your recording and compares two forms of spoken assistance. In gaze-contingent blocks, coarse directional guidance helped you orient toward the target area before closer/farther feedback narrowed the location. In noncontingent blocks, general prompts occurred at matching opportunities but did not use gaze, hover, coverage, or target-proximity information.

There was no deception in the planned procedure. Responsiveness differed because gaze contingency was an experimental factor. The self-similar voice was created only for fixed study prompts and could produce those phrases even though you had not recorded them word-for-word.

Eye-gaze measures are interaction telemetry, not a medical test; the study does not diagnose vision, attention, or neurological conditions. Hints or gaze estimates may be delayed or incorrect.

1.0 Voice and Data Cleanup

The team will follow the approved procedure for transferring coded study data and deleting the local recording, private provider voice/sample, generated clips/history where supported, and headset speech cache. Provider backup, security-log, or legal-retention limits are described in your consent form.

If you want to withdraw or ask whether identifiable voice assets can still be deleted, contact the Principal Investigator at **[OPEN DECISION: name and UCF contact]**. Information already de-identified or combined with other participants' results may not be retrievable.

2.0 After the Headset

Tell the researcher before leaving if you have nausea, dizziness, headache, eyestrain, blurred vision, disorientation, balance difficulty, neck/back discomfort, or another concern. Avoid driving or risky activity until you feel fully normal; seek medical help if symptoms are severe, worsening, or persistent.

3.0 Contacts

Research questions, problems, or withdrawal/deletion requests: **[OPEN DECISION: PI name, phone, UCF email]**

Faculty advisor, if applicable: **[OPEN DECISION]**

Questions about participant rights or concerns independent of the study team: UCF Institutional Review Board, 407-823-2901, irb@ucf.edu. **[Confirm approved wording.]**

If you would like an aggregate summary when the study is complete: **[OPEN DECISION: approved opt-in method that does not unnecessarily link contact details to study data.]**

UCF IRB study number: **[add after assignment/approval]**

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